THE FLATNESS PROPERTY P IS STABLE UNDER COMPOSITION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: P is defined (not proved) by lemma-flat-axioms, so this card does not depend on the rejected clause (3) of that lemma's proof; the Formal Spaces remark reference in the statement is statement-side

Editorial note. The label lemma-composition-flat-continuous names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Lemma lemma-flat-axioms (in-source definition carrier): defines the property P of arrows of WAdm^Noeth by flatness of the relevant localizations
- standard ring facts: localizations of flat ring maps are flat, and flat ring maps are stable under composition (Algebra, commutative algebra facts)

Source reference. Original remark 0GBK: sources/stacks-chow/tags.txt, source lines 20375–20375. Prerequisite pointer invoked at 3015; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0GC9: sources/stacks-restricted/pinned-original-source.tex, source lines 2890–2890. Prerequisite pointer invoked at 3013; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Denote P the property of arrows of $WAdm^{Noeth}$ defined in Lemma 0GC9. Then P is stable under composition (Formal Spaces, Remark 0GBK).*

Proof. This is true because compositions of flat ring maps are flat. □

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